MAH QING TONG

LLM ENGINEER | AI ENGINEER

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PROFESSIONAL SUMMARY

LLM Engineer at Zetrix AI with a Bachelor of Computer Science in Data Science from **Monash University Malaysia** (Second Upper Honours). Experienced in architecting production Agentic RAG workflows, modular **Agent Skills (`SKILL.md` / MCP)**, and multi-agent graphs with practical harness engineering. Frequently serving as Person-in-Charge (PIC) across key AI initiatives, taking full end-to-end technical responsibility from initial system design to production delivery.

TECHNICAL SKILLS

LLM & Agent Systems:	LangGraph, LangChain, Agent Skills (SKILL.md), Model Context Protocol (MCP), Dify, Multi-Agent Orchestration, Human-in-the-Loop (HITL), Prompt Engineering
RAG & Retrieval:	Hybrid Search, Dense & Sparse Retrieval (BM25), Qdrant, ChromaDB, Cross-Encoder Reranking, Multi-Hop Retrieval, Document Preprocessing
Programming & Backend:	Python, FastAPI, TypeScript, JavaScript, SQL, REST APIs, Next.js, Server-Sent Events (SSE)
Machine Learning & DevOps:	PyTorch, Hugging Face Transformers, BERT Prefix Prompt-Tuning, Docker, Linux, Git, PostgreSQL, DuckDB, CI/CD

TECHNICAL PROJECTS

Personal DeepAgent — AI Agent System & Kinetic Portfolio `LangGraph` • `Agent Skills` • `Hybrid RAG` • `FastAPI` • `DuckDB` • `FlashRank` • `Next.js` • `SSE`

- Architected a multi-agent orchestration graph featuring dynamic planning with step-budget guardrails, a ReAct research agent, and modular sub-agent workers with isolated execution.
- Implemented modular Agent Skills (`SKILL.md` / MCP) and two-stage hybrid retrieval combining dense vector embeddings with sparse BM25 indexing in DuckDB and FlashRank cross-encoder reranking.
- Engineered deterministic tool execution for system commands, multi-engine web search, and file operations with automated failure recovery and self-correction loops.
- Integrated a full-stack kinetic portfolio web interface (Next.js, Tailwind CSS) powered by an asynchronous FastAPI backend with Server-Sent Events (SSE) streaming and Human-in-the-Loop approval workflows.

AI Job Search & Resume Agent `LangChain` • `MCP` • `Qdrant` • `Hybrid Search` • `JSearch API`

- Designed a multi-agent LLM application automating job discovery, semantic candidate ranking, resume analysis, and personalised outreach drafting.
- Implemented dense and sparse BM25 hybrid retrieval with cross-encoder reranking in Qdrant for high-precision resume-to-job matching.
- Developed an orchestration workflow managing query rewriting, information retrieval, ranking, and final response generation.

PROFESSIONAL EXPERIENCE

Zetrix AI | LLM Engineer — JUN 2026 – PRESENT

- **Enterprise AI Workflow Automation:** Developed Dify-based AI workflows for enterprise SOP and knowledge-base retrieval, grounding LLM responses in internal documentation to improve answer relevance and consistency for customer operations.
- **Section-Aware RAG Ingestion:** Built automated preprocessing pipelines converting unstructured PDF research into hierarchical Markdown before chunking, improving downstream retrieval recall by 28% in measured evaluations.
- **Agentic RAG & Deterministic Execution:** Developed the i-FIKR Islamic knowledge agent over 870+ records, integrating vector search with custom Python and Pandas tools for accurate multi-criteria filtering and tally operations.
- **Multi-Hop Retrieval & Citation Grounding:** Implemented a two-stage retrieval workflow combining catalog-level metadata grounding with targeted document retrieval, enabling cross-document comparisons with verifiable sub-chunk citations.

NIQ (NielsenIQ) | Data Input Validation Intern — NOV 2024 – FEB 2025

- **Large-Scale Dataset Validation:** Processed and validated retail datasets containing thousands of records per batch, ensuring data consistency and integrity for downstream analytics.
- **Rule-Based Anomaly Detection:** Applied logic-driven validation techniques to identify catalog inconsistencies across regional data pipelines (Malaysia & Vietnam).

EDUCATION

Monash University Malaysia | Bachelor of Computer Science in Data Science — FEB 2023 – DEC 2025

Second Upper Honours • CGPA: 3.36 / 4.00

- **Deep Learning & Neural Architectures:** Implemented Multi-Head Self-Attention modules, Position-wise Feed-Forward networks, and Transformer Encoders from scratch in PyTorch; architected Prefix Prompt-Tuning pipelines for BERT model adaptation.